



Nawatech

Beyond Engineering

nawatech.co



KAI Workshop – Day 7

Prepared for: PT Kereta Api

Indonesia

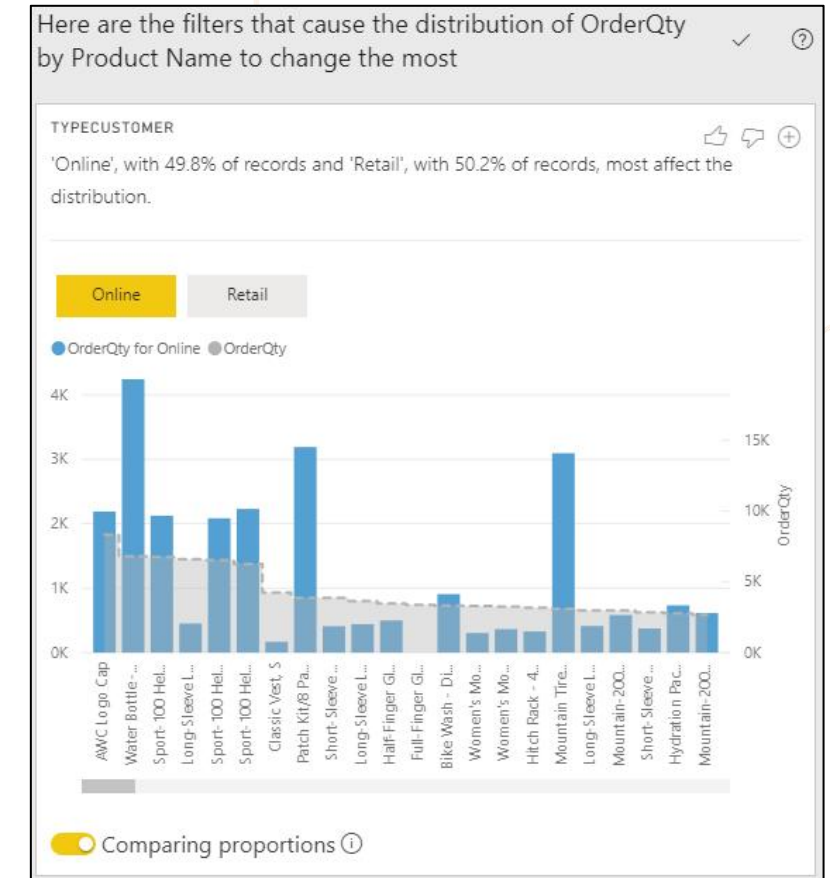
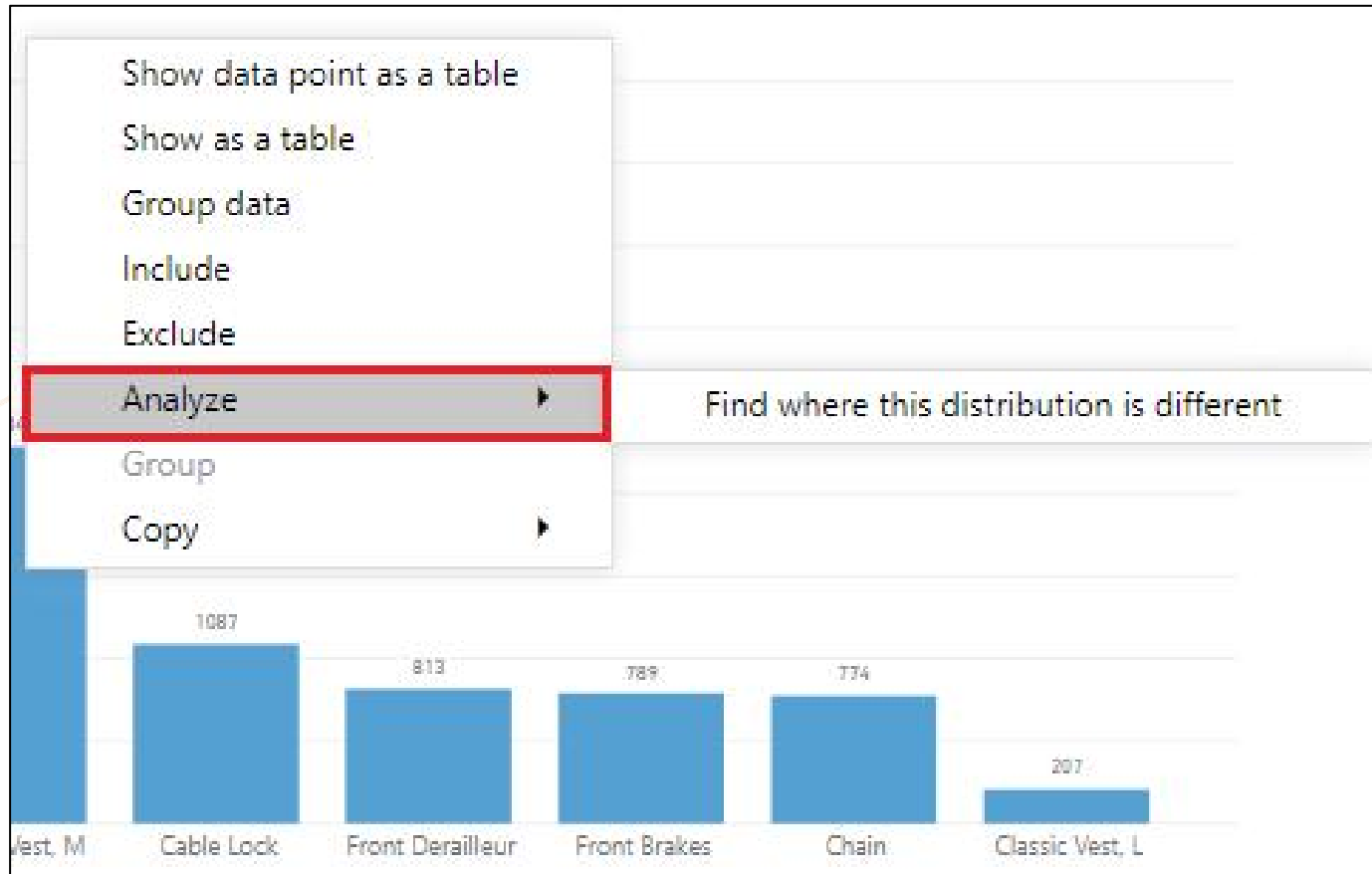


Perform advanced analytics





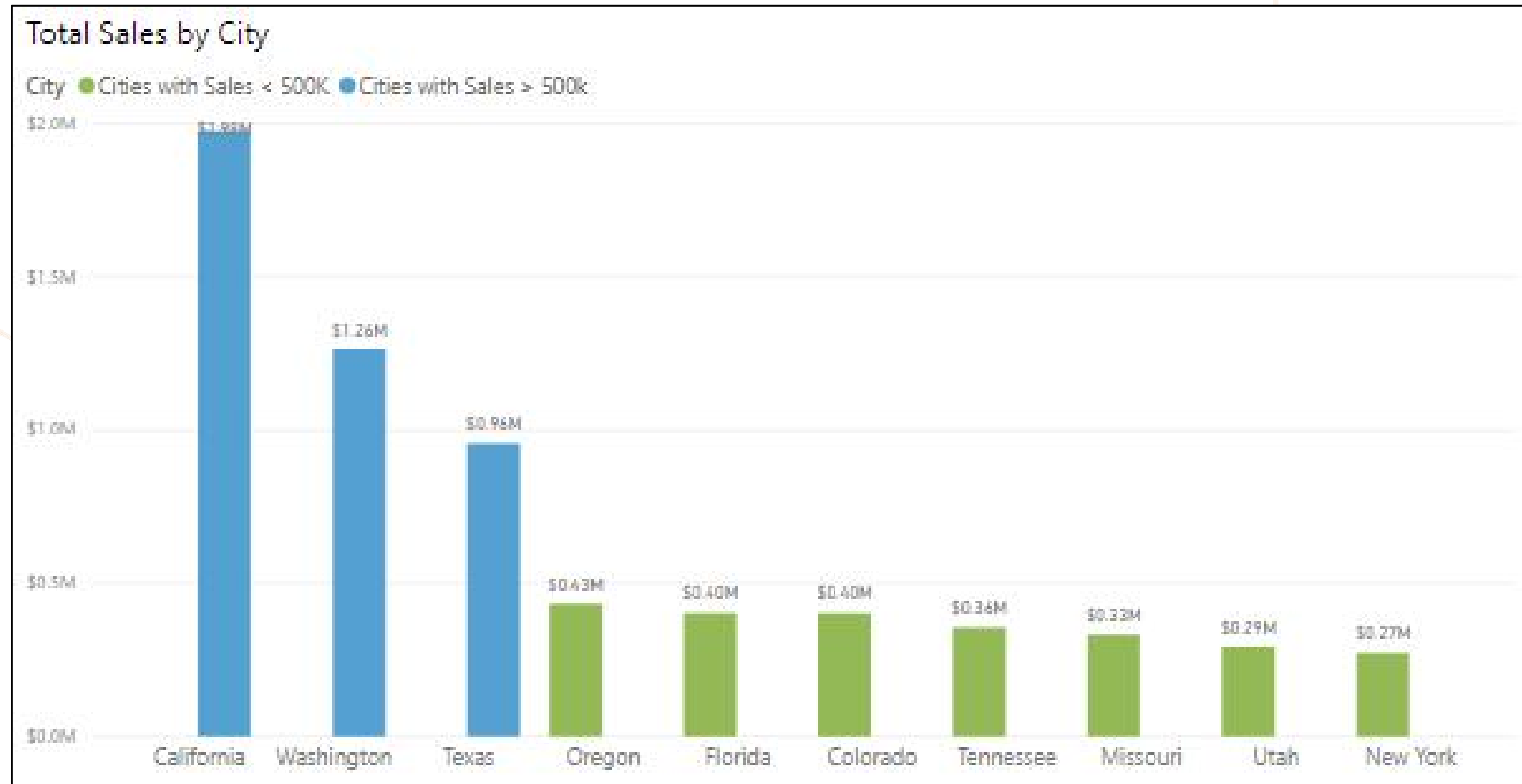
Compare data with the Analyze feature





Group and bin data

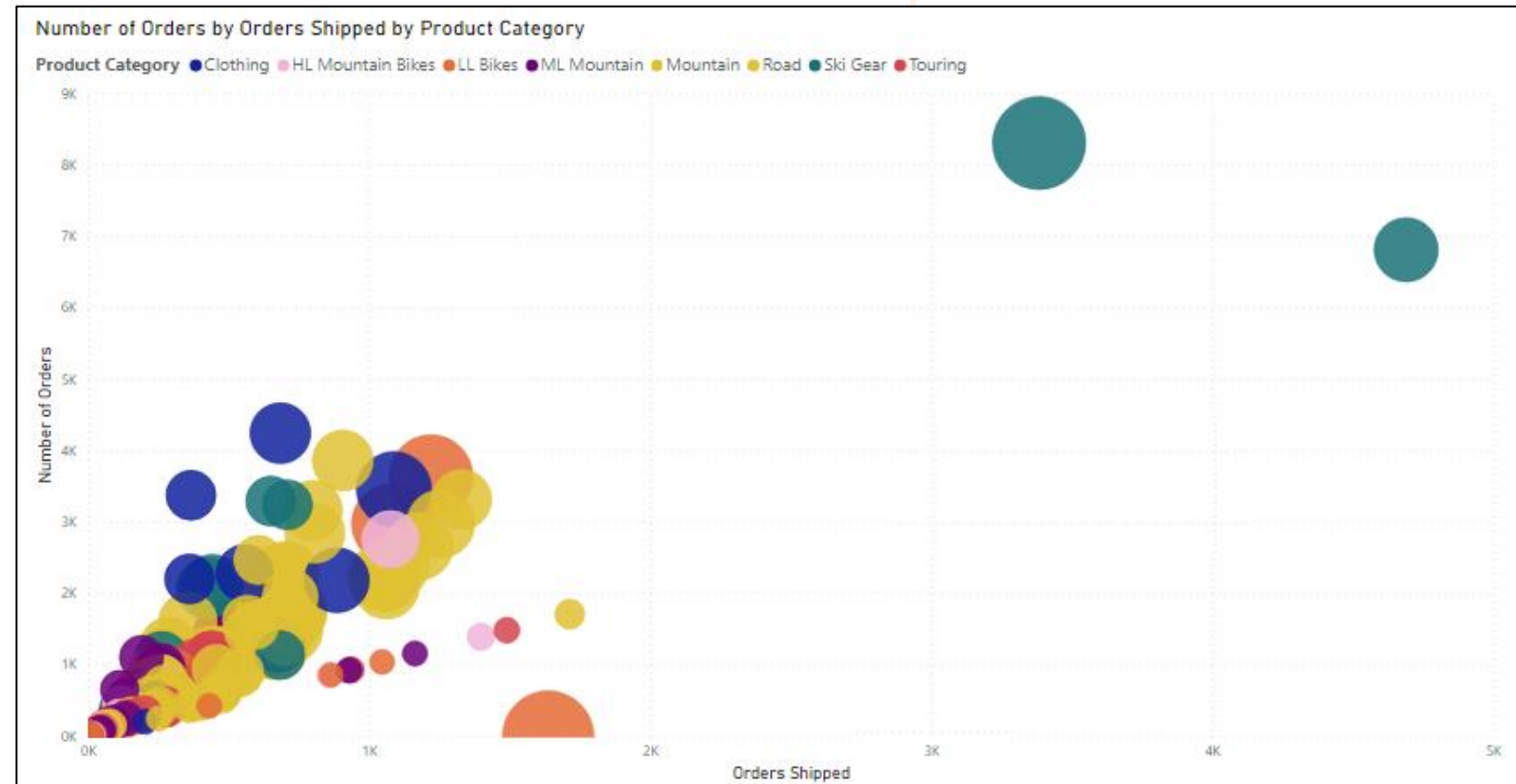
- Organize data into categories or intervals to better analyze and visual data.





Identify outliers in data

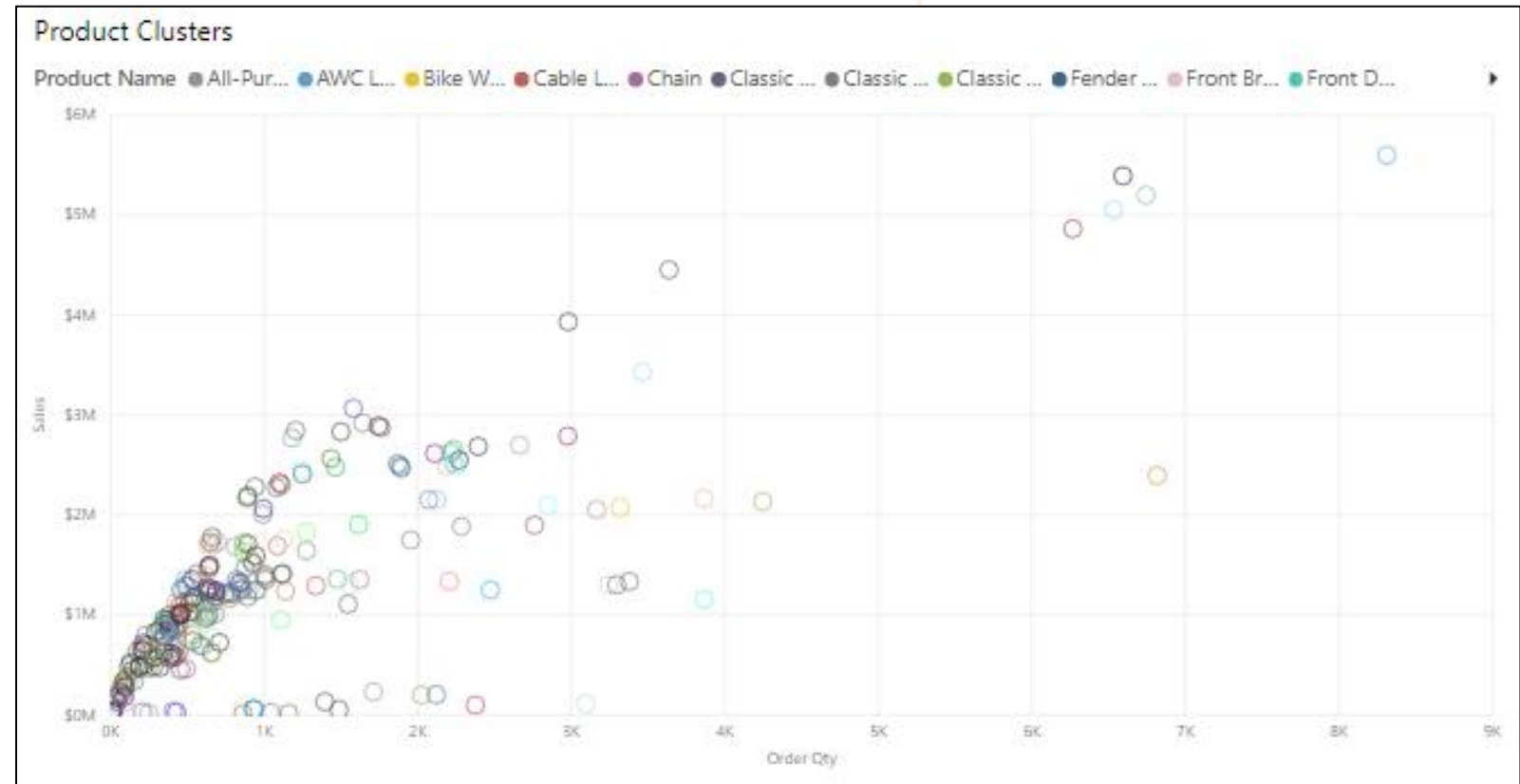
- Unexpected deviations in data.
- Scatter charts can detect outliers.
- Use DAX to define and analyze outliers dynamically.





Clustering with Scatter chart visual

- Automatic data grouping.
- Enhance pattern recognition and insights.
- Simplify data understanding and analysis.





Key Performance Indicator visual

- At-a-glance insights into performance against predefined targets.



Value

Add data fields here

Trend axis

Add data fields here

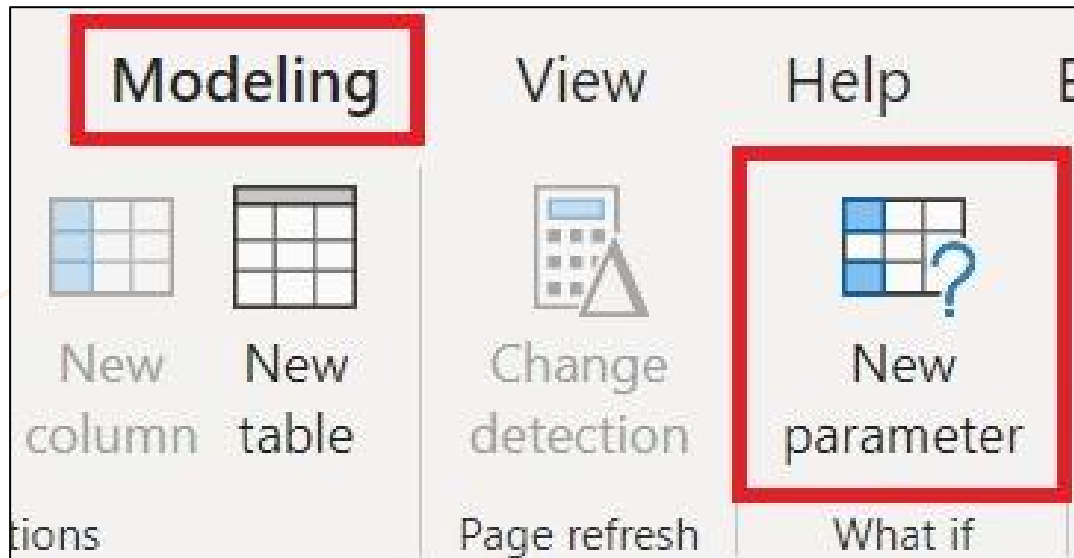
Target

Add data fields here



What-if parameters

- Enable scenario planning and interactive exploration for better decision-making.



What-if parameter

Name

Data type

Minimum

Maximum

Increment

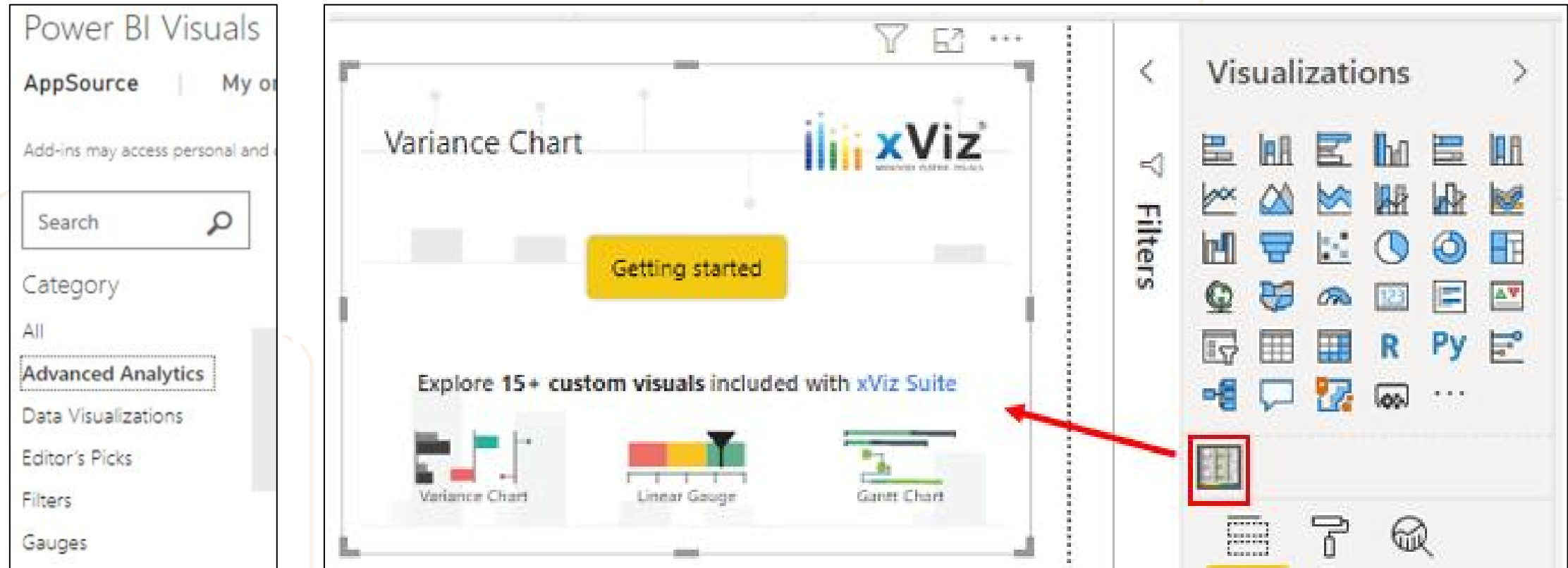
Default

☒ Add slicer to this page



Advanced Analytics custom visuals

- Discover additional visuals developed with Advanced Analytics in mind.





AI Insights for data preparation in Power Query Editor

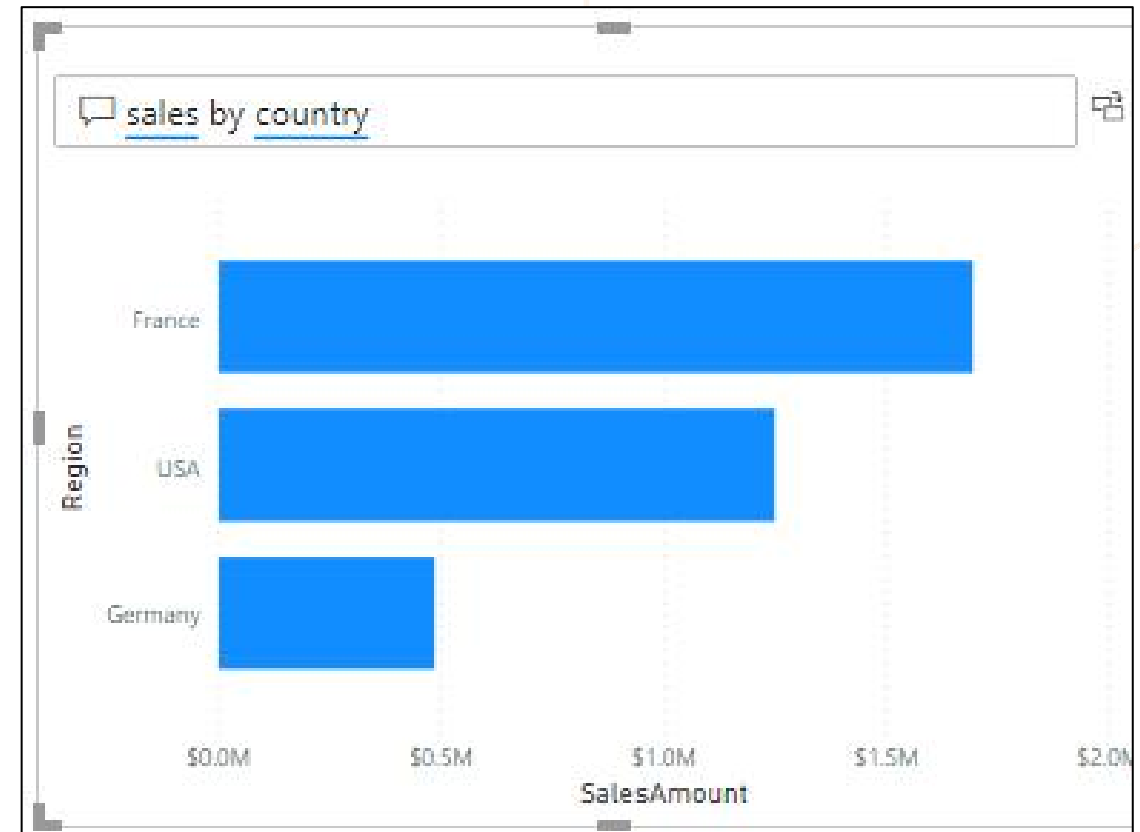
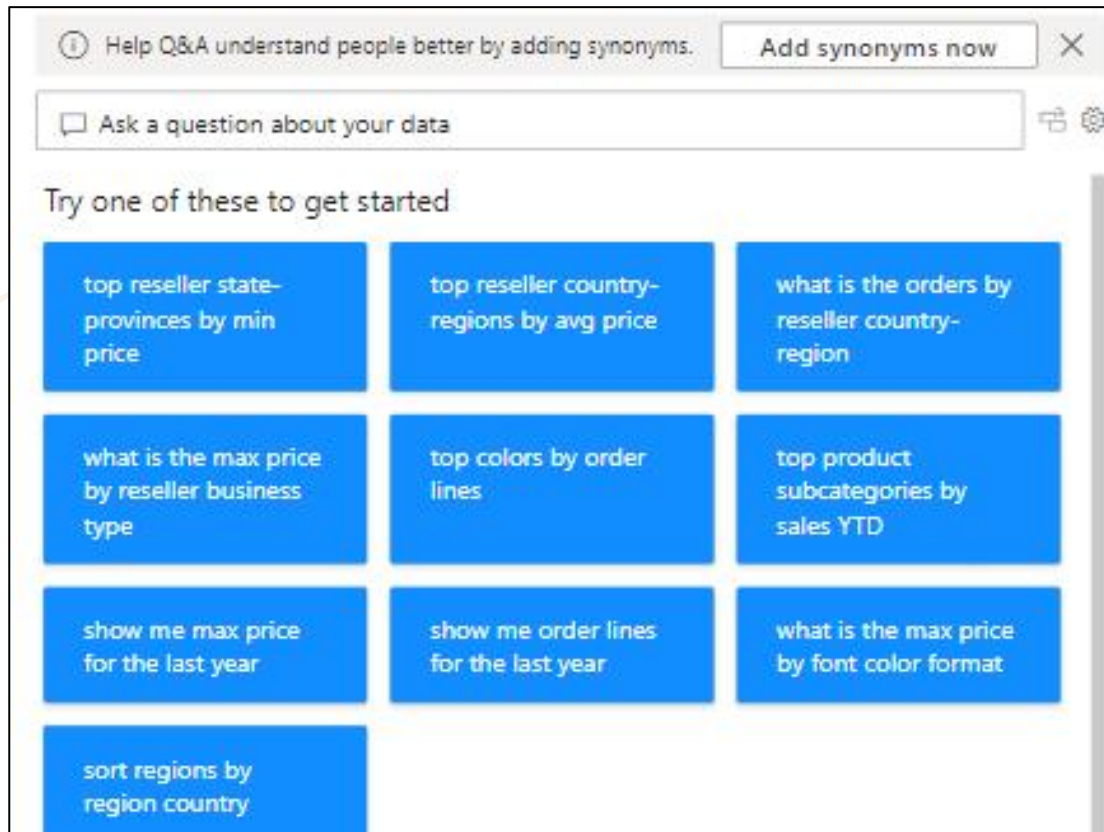
- Leverage AI analysis of text, images, and use Azure Machine Learning models.





AI in the Q&A feature

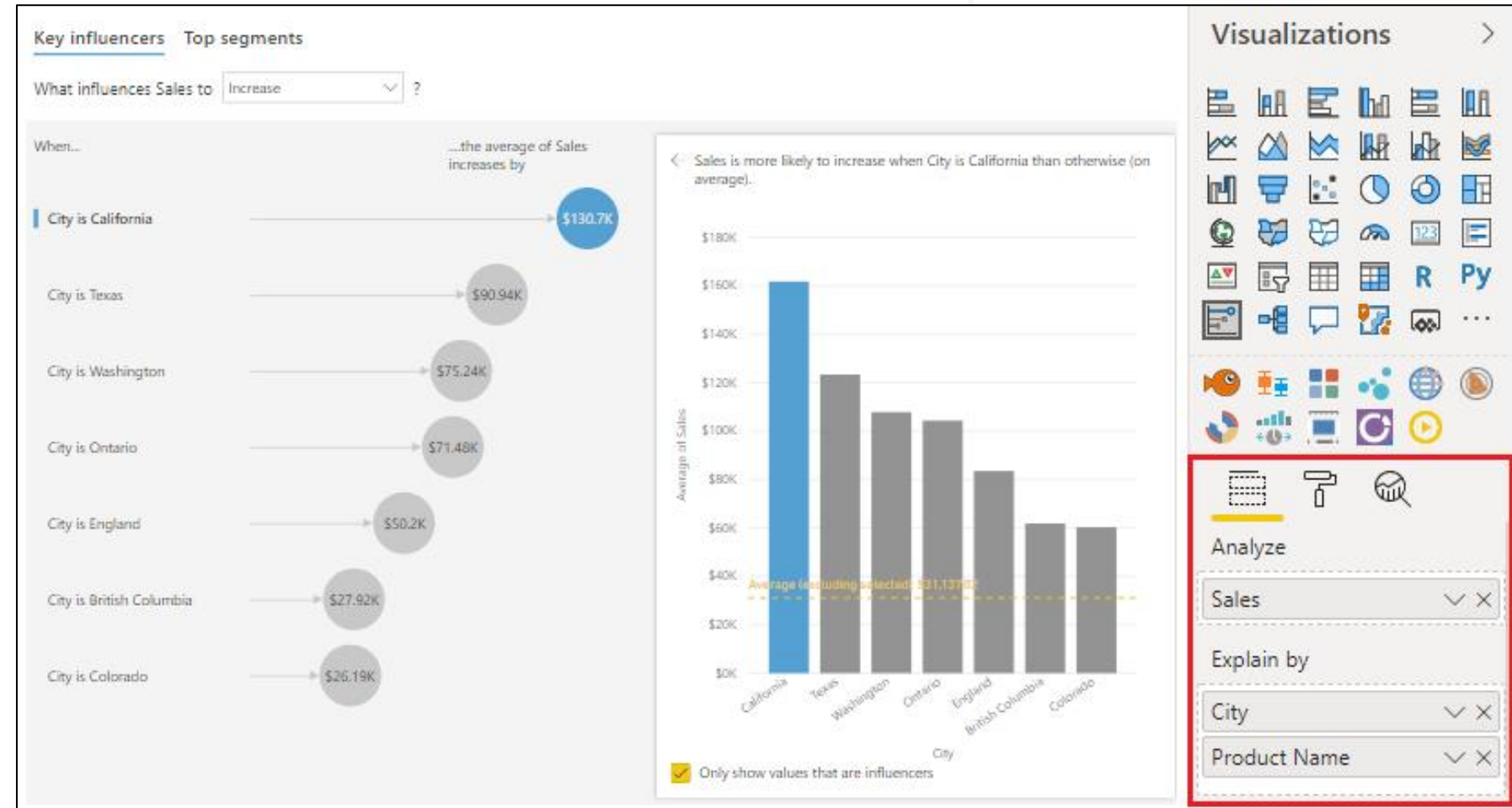
- Use natural language to ask questions about data and create visuals for insights.





Key influencers visual

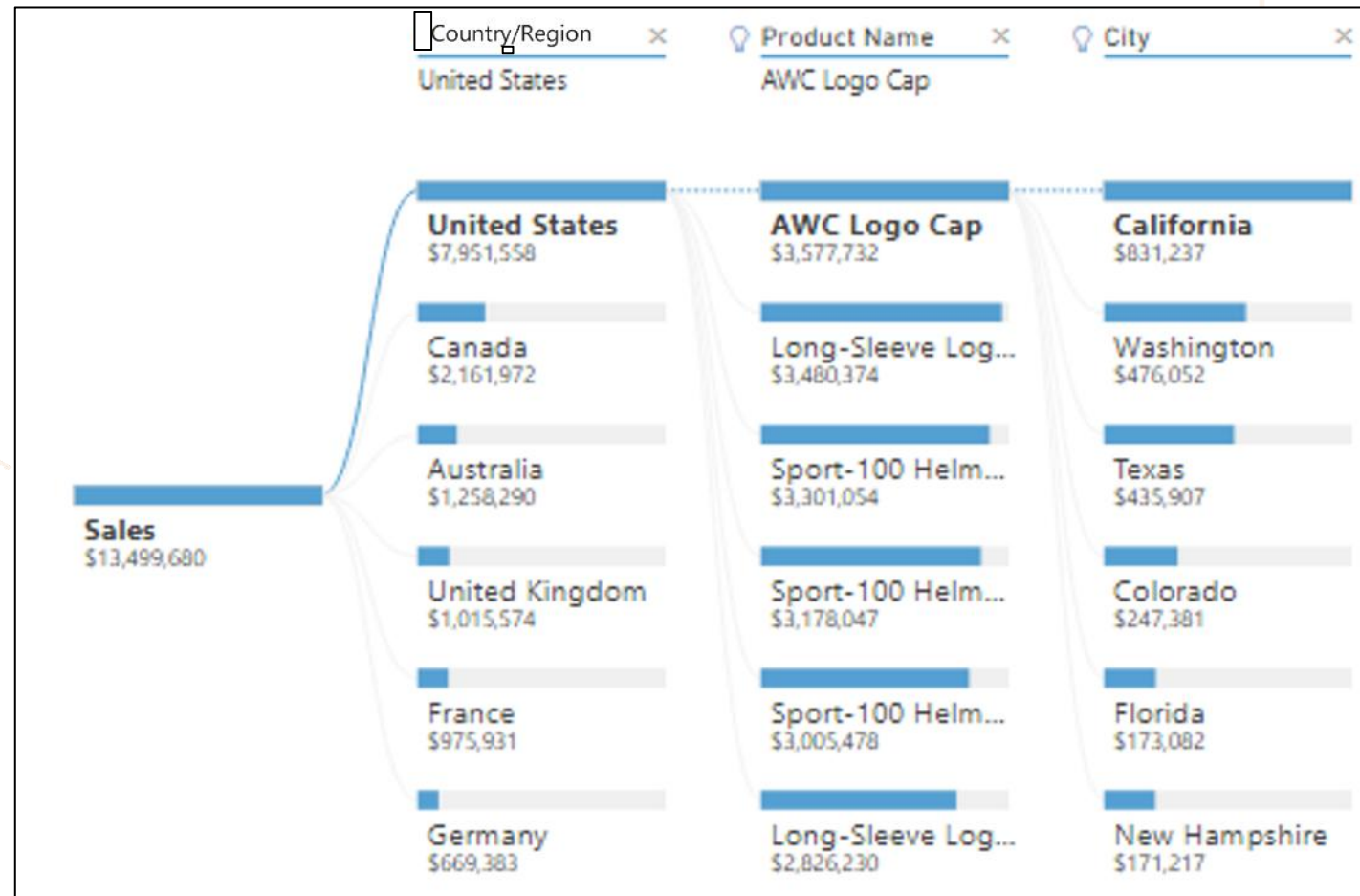
- Identify factors driving variations in a chosen target metric.





Decomposition tree visual

- Visualize data hierarchies for in-depth exploration and pattern discovery.

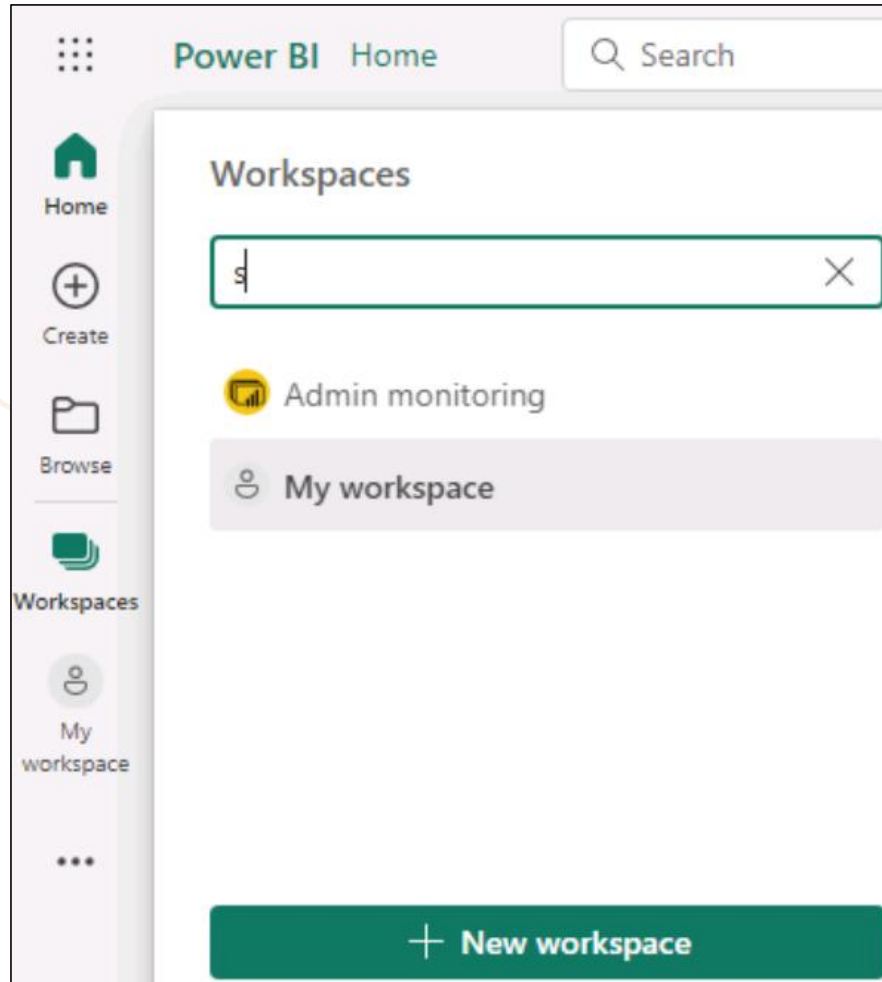


Create and manage workspaces





Create Power BI workspaces



Create a workspace

Name *

Available

Description

Domain (preview) ⓘ

[Learn more about workspace settings](#)

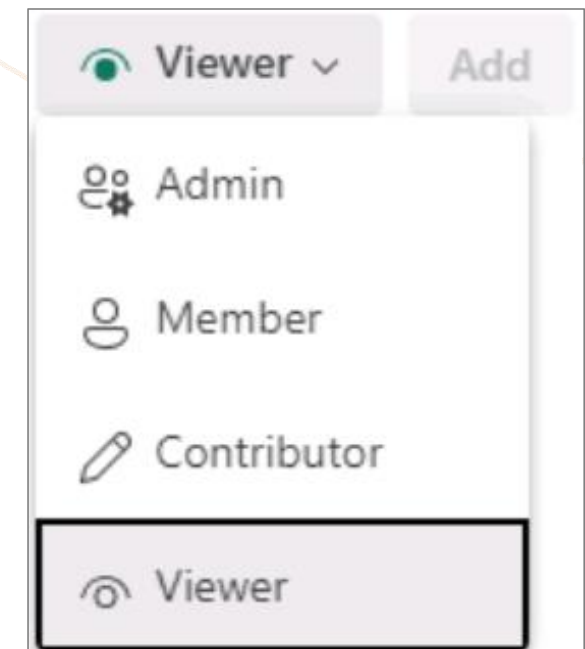
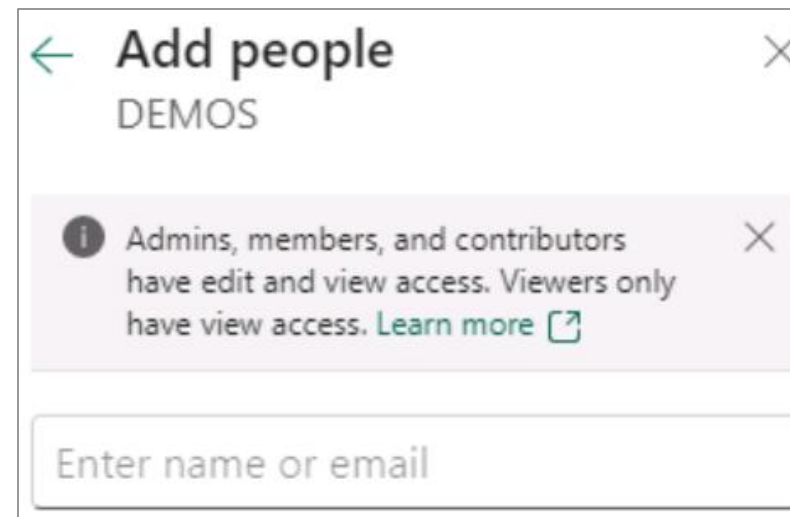
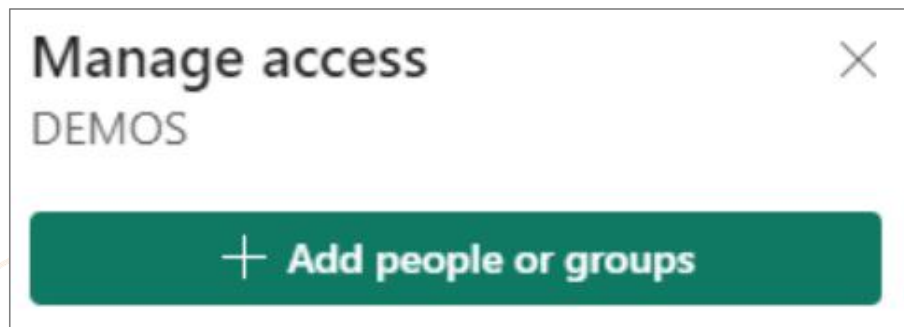
Workspace image

Advanced ▼



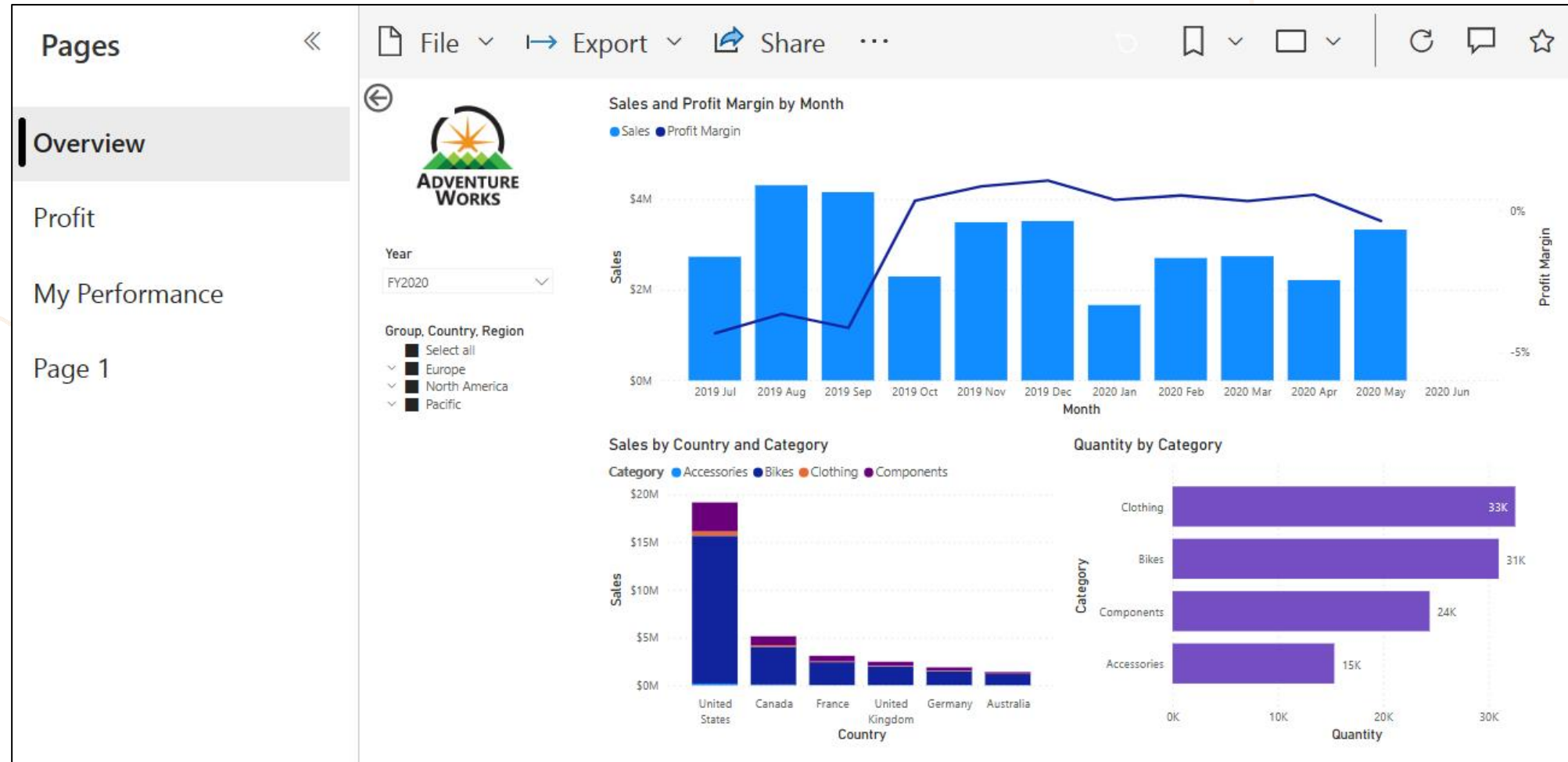
Assign workspace roles

- Use workspaces for collaboration with other developers, not content sharing.



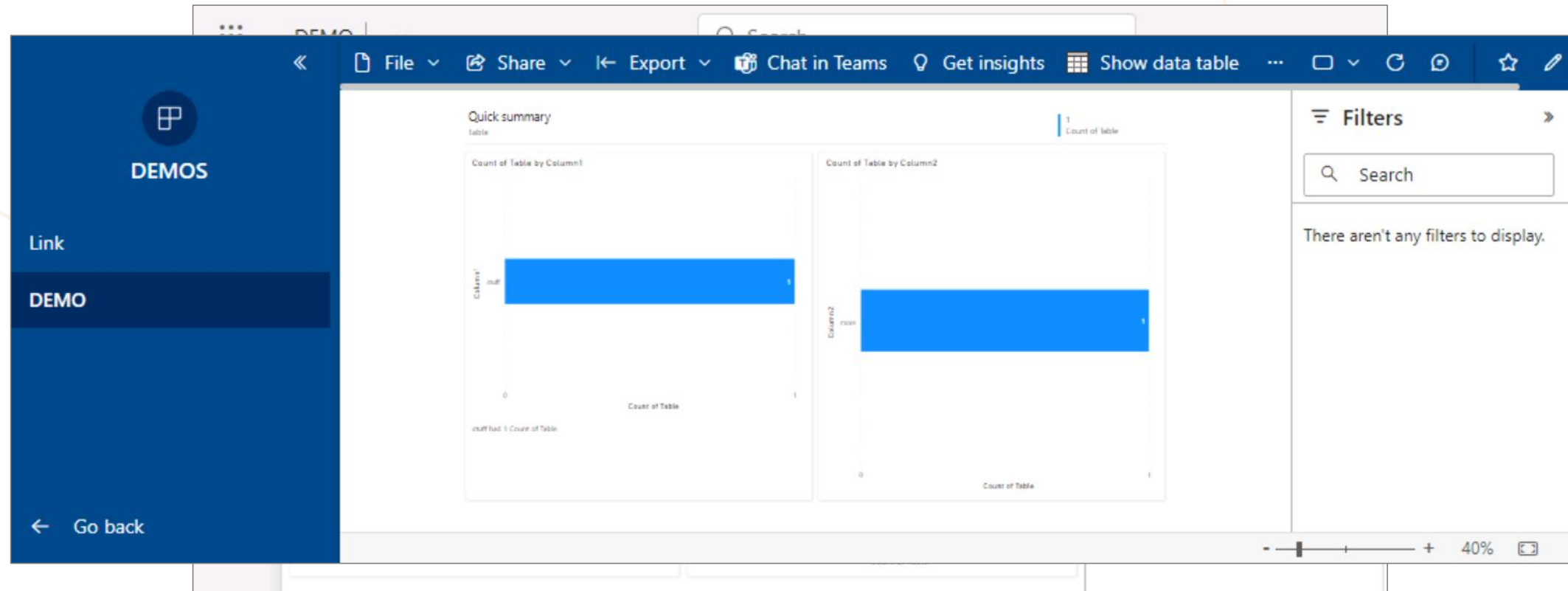


Publish reports to a workspace



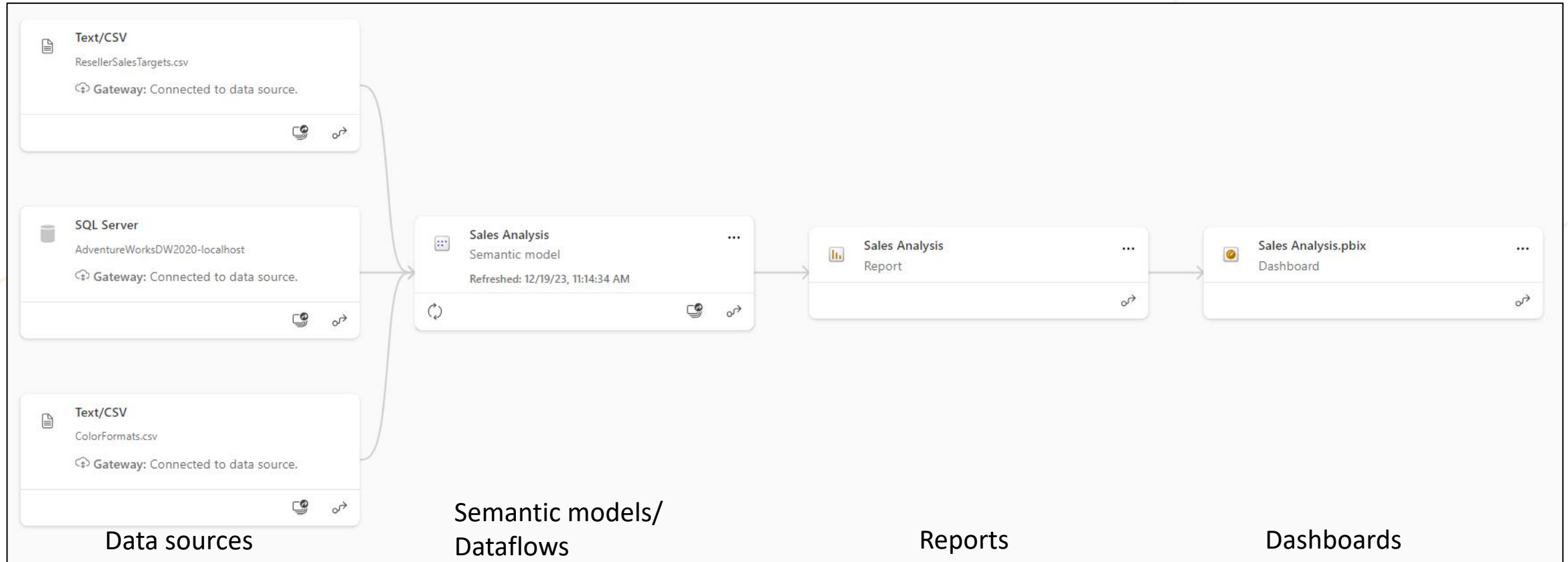


Explore different workspace items





Identify relationships between items with Lineage view





Configure apps for distribution

① Setup ② Content ③ Audience

Create app

Audience

Manage

Success

DEMONS

Give people

<https://app>

Build your app

App name *

DEMOS

Description *

Enter a summary

Describe your app.

App logo

Upload

Delete

App theme color

Global App Settings

Install this app automatically.

Hide app navigation pane.

Allow users to make a copy of the reports in this app.

Support site

Share where your users can find help

Contact Information

Show app publisher

Show items contacts from the workspace

Show specific individuals or groups

Enter a name or email address

Copy

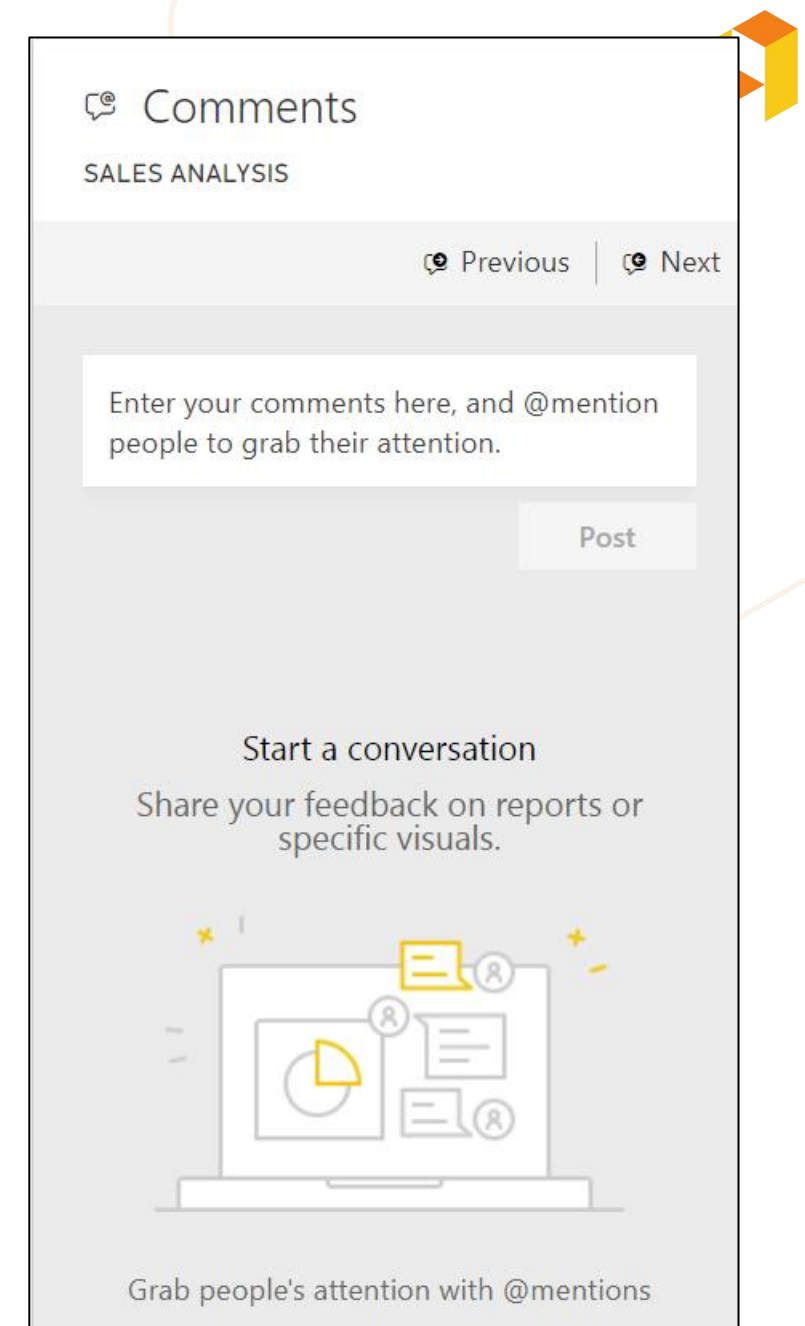
to app

Close

Workspace users

Provide feedback in report comments

- Comment pane
- Tag specific people
- Comment on specific visuals





Power BI licensing overview

License	Shared workspace	Premium workspace
Free	• Access to their own content. • Cannot distribute any content.	• Access workspace content. • Cannot distribute any content.
Pro	• Access to all workspace content. • Distribute to other Pro users.	• Access all workspace content. • Distribute content to all users.
Premium Per User (PPU)	• Access to all workspace content. • Distribute to other PPU users.	• Access all workspace content. • Distribute content to all users.

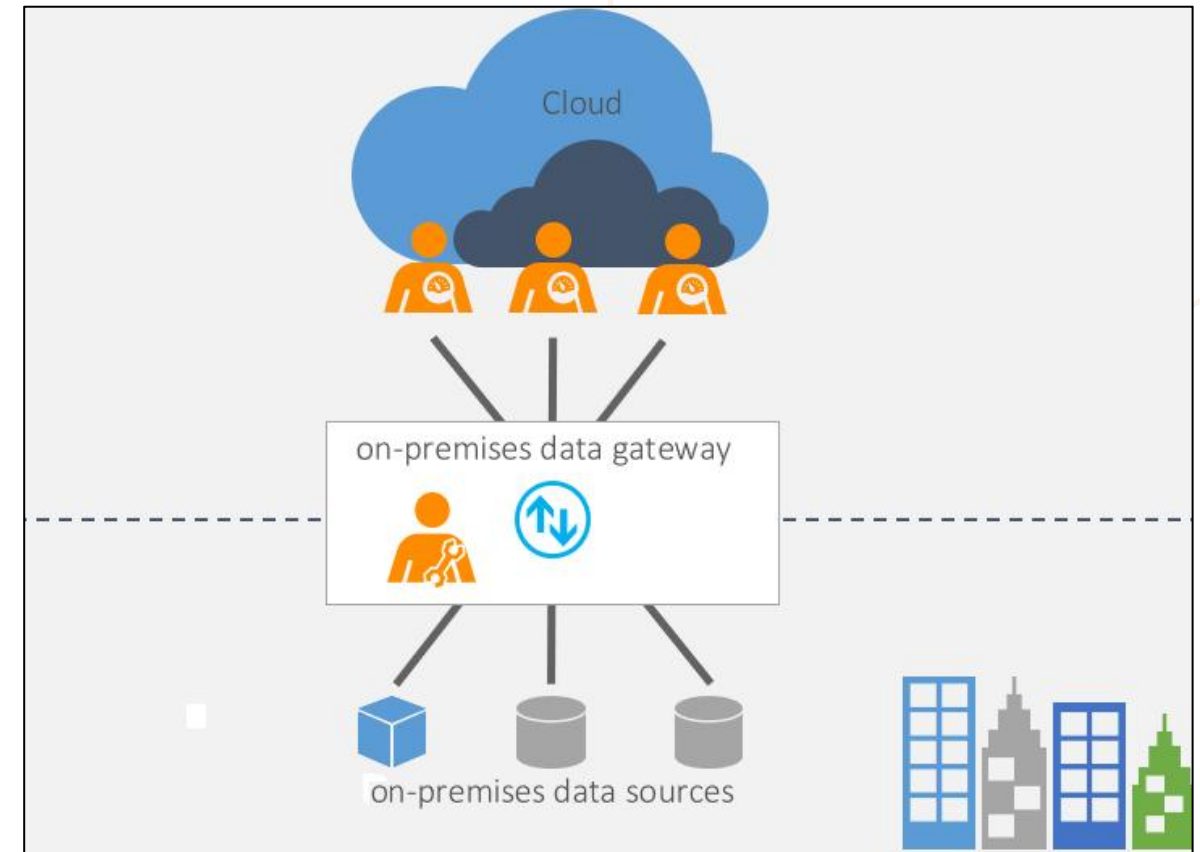
Manage semantic models





Connect to on-premises data sources

- Gateways allow Power BI to connect and refresh **on-premises data sources.**
- Organizational or personal gateways available.
- Require credentials to data source.





Keep data up to date

On-demand or scheduled refresh

- 8 times per day for shared
- Up to 48 times per day with Premium

Automatically pauses after

- 5 consecutive failures
- 2 months of inactivity

Scheduled refresh

Keep your data up to date

Configure a data refresh schedule to import data from the data source into the dataset.

☒ On

Refresh frequency

Daily ▼

Time zone

(UTC-08:00) Pacific Time (US and Can: ▼

Time

[Add another time](#)

Send refresh failure notifications to

☒ Dataset owner

☐ These contacts:

Enter email addresses

Apply

Discard



Incremental refresh

- Works best with query folding
- DirectQuery requires Premium/PPU

How to enable:

- Define the filter parameters
- Use the parameters to apply a filter
- Define the incremental refresh policy

Incremental refresh and real-time data

Refresh large tables faster with incremental refresh. Plus, get the latest data in real time with DirectQuery (Premium only). [Learn more](#)

 These settings will apply when you publish the dataset to the Power BI service. Once you do that, you won't be able to download it back to Power BI Desktop. [Learn more](#)

1. Select table

Orders 

2. Set import and refresh ranges

☒ Incrementally refresh this table

Archive data starting  Years before refresh date

Data imported from 1/1/2017 to 11/17/2022 (inclusive)

Incrementally refresh data starting  Days before refresh date

Data will be incrementally refreshed from 11/18/2022 to 11/20/2022 (inclusive)

3. Choose optional settings

☐ Get the latest data in real time with DirectQuery (Premium only) [Learn more](#)

Selected table cannot be folded for DirectQuery.

☐ Only refresh complete days [Learn more](#)

☐ Detect data changes [Learn more](#)

4. Review and apply

Archived

Incremental Refresh



Semantic model endorsement

- Provide trusted semantic models and encourage reusability.

Endorsement and discovery

Help coworkers find your quality content by endorsing this semantic model and making it discoverable. [Learn more](#)

☒ None
This semantic model will appear in search results but isn't endorsed.

☐ Promoted
When you're ready to distribute the semantic model to your coworkers, promote it to let them know.

☐ Certified
Certify your semantic model to show coworkers that it's been reviewed and meets your org's certification criteria. [How do I get my semantic model certified?](#)

☐ Make discoverable
Allow users without access to this semantic model to discover it and request permissions to access the data [Learn more](#)

Apply

Discard



Data protection report settings

- If and how users can export data.

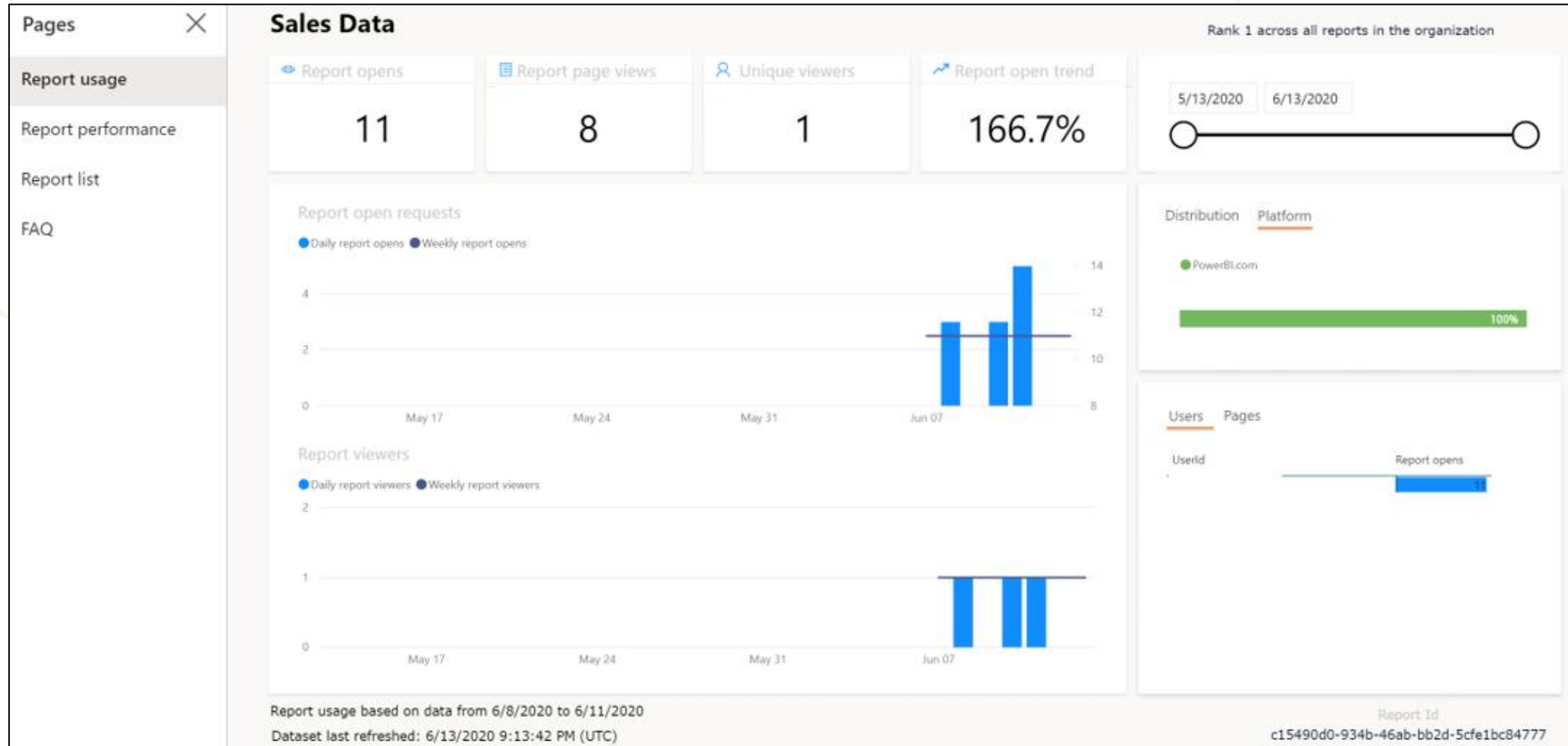
Export data
Choose the type of data you allow your end users to export.
Summarized data and data with current layout
Summarized data and data with current layout
Summarized data, data with current layout and underlying data
None

- Set data sensitivity.

Sensitivity label *
Classify the sensitivity of this report content.
Non-Business
Non-Business
Public
General



Leverage Usage Metrics report for insights



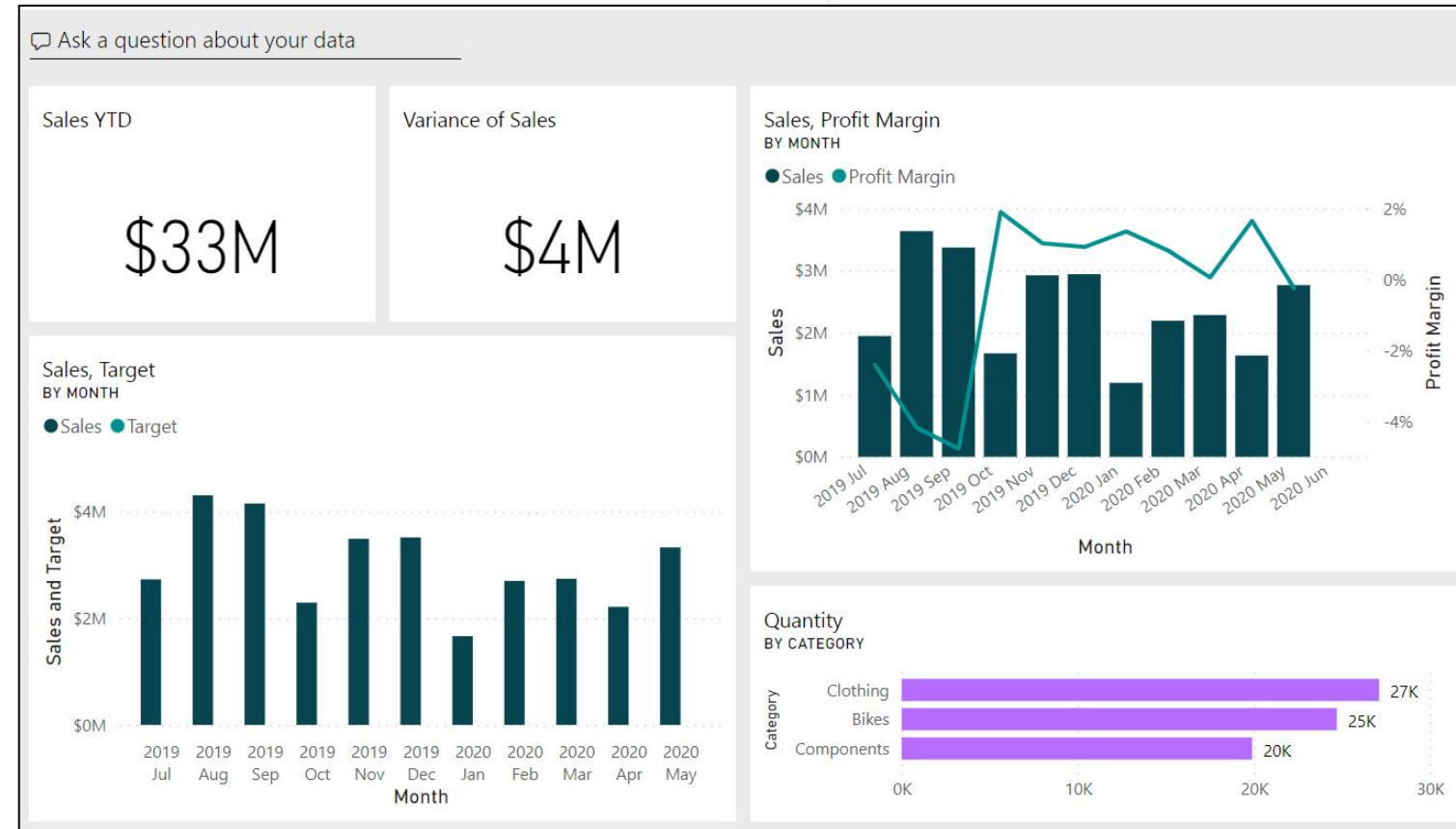
Create dashboards





What is a Power BI dashboard?

- Single page
- Visuals pinned from reports
- Tiles are static
- Default Q&A feature
- Refreshes with semantic model
 - DirectQuery every 15 min





Pin visuals to a dashboard

- Specific visuals
- Entire pages (pin live)

Pin to dashboard

Select an existing dashboard or create a new one.

Where would you like to pin to?

☒ Existing dashboard

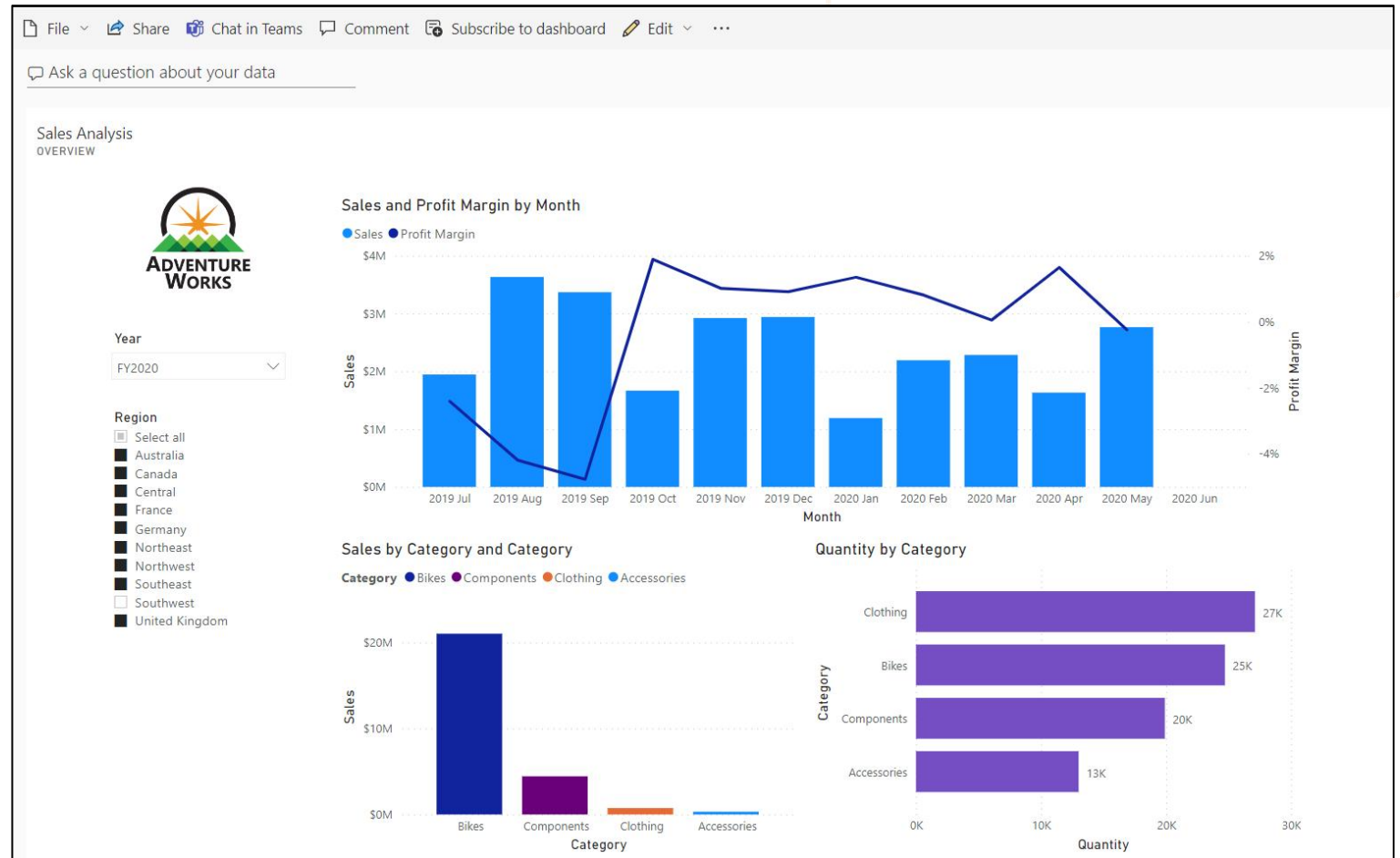
☐ New dashboard

Select existing dashboard

Sales Overview

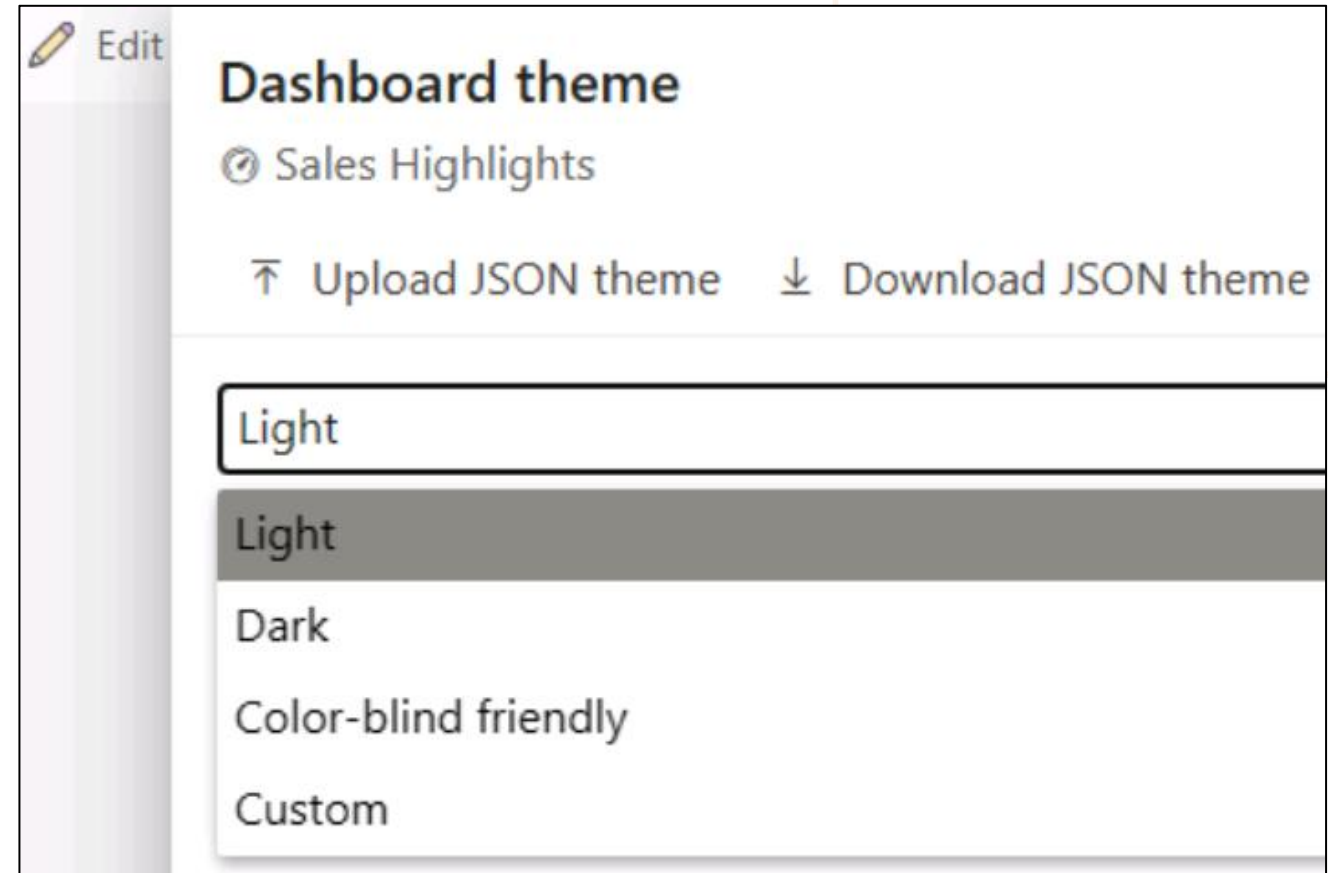
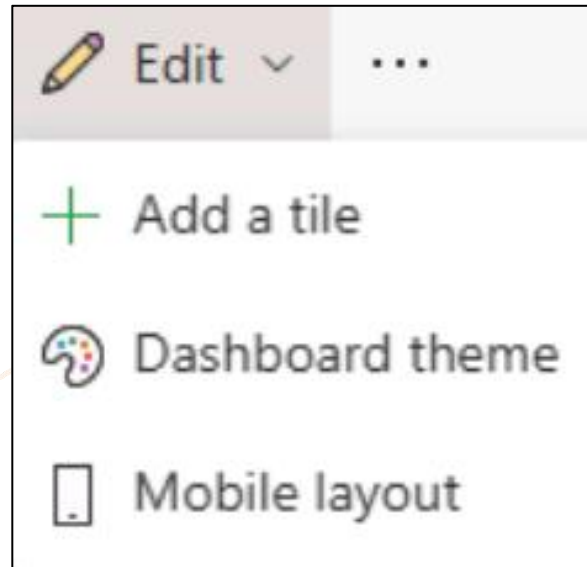
Pin live page enables changes to reports to appear in the dashboard tile when the page is refreshed.

Pin live Cancel



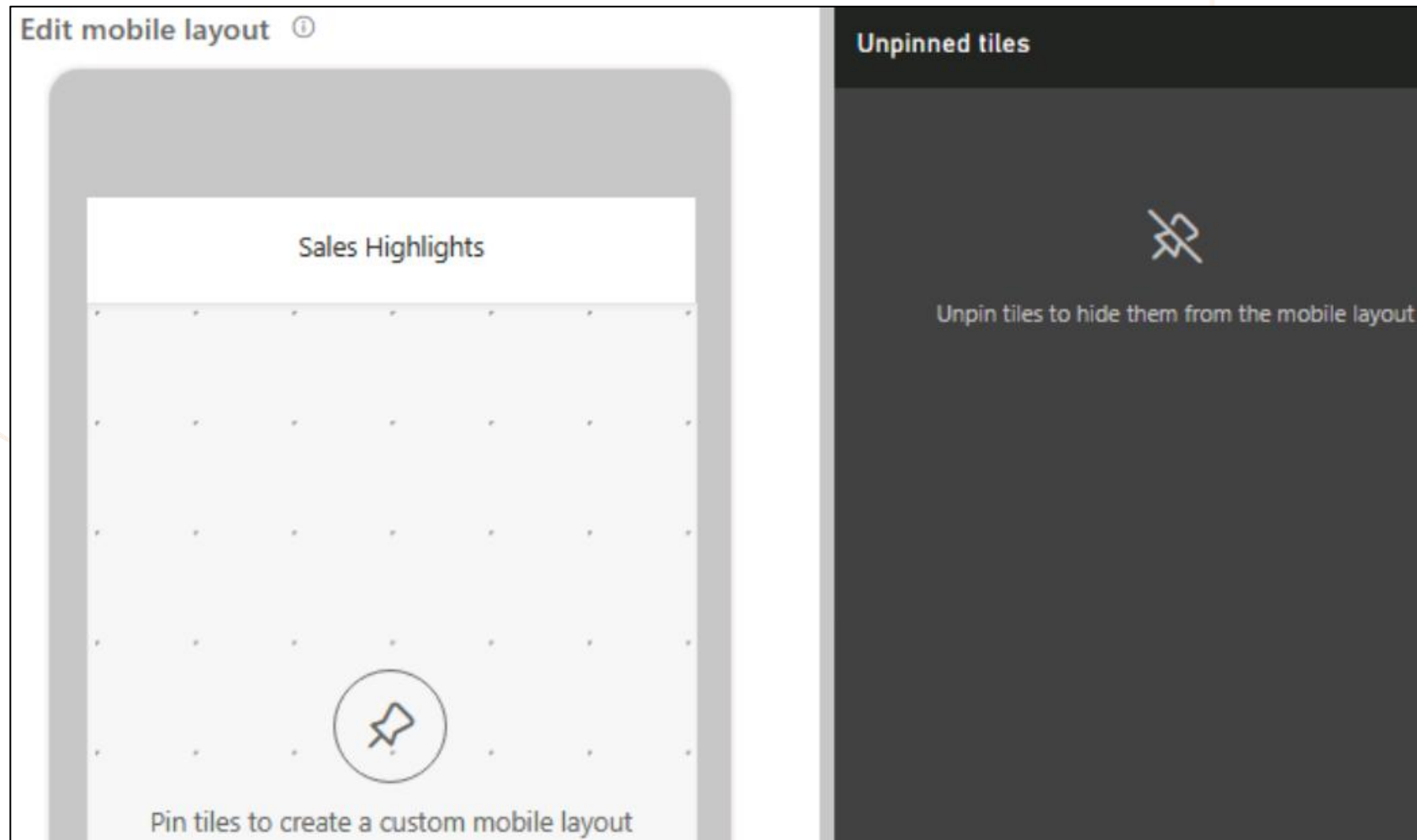


Enhance your dashboard with a theme



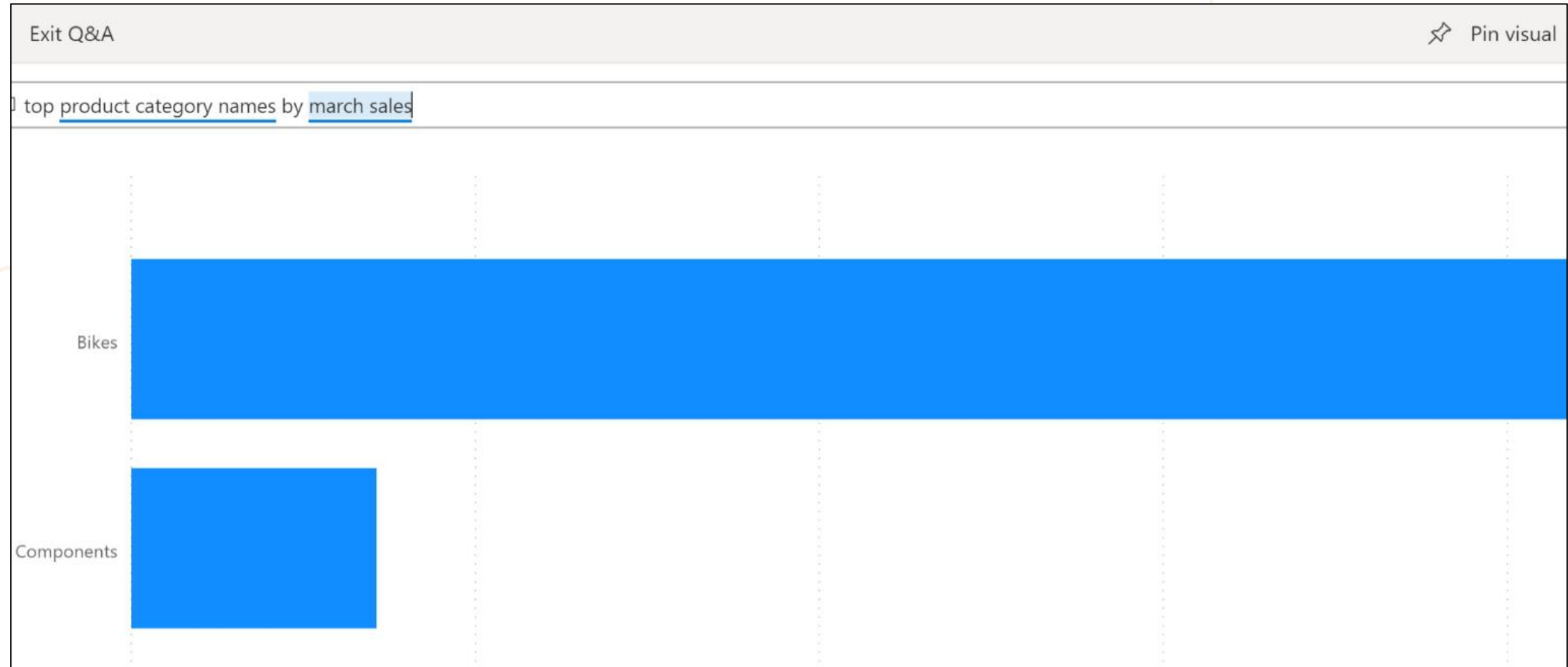


Design mobile layout for dashboards on the go





Use natural language to explore with Q&A





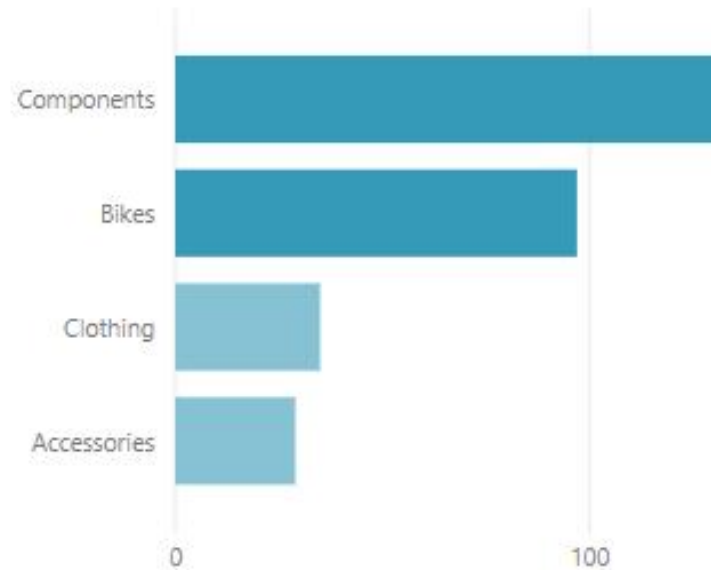
Discover more with Quick Insights

- Uses Machine Learning algorithms.
- Generate visual interactions.
- Find insights missed during development.

Quick Insights

A subset of your data was analyzed and the following insights were found. [Learn more](#)

Count of Product
BY CATEGORY NAME



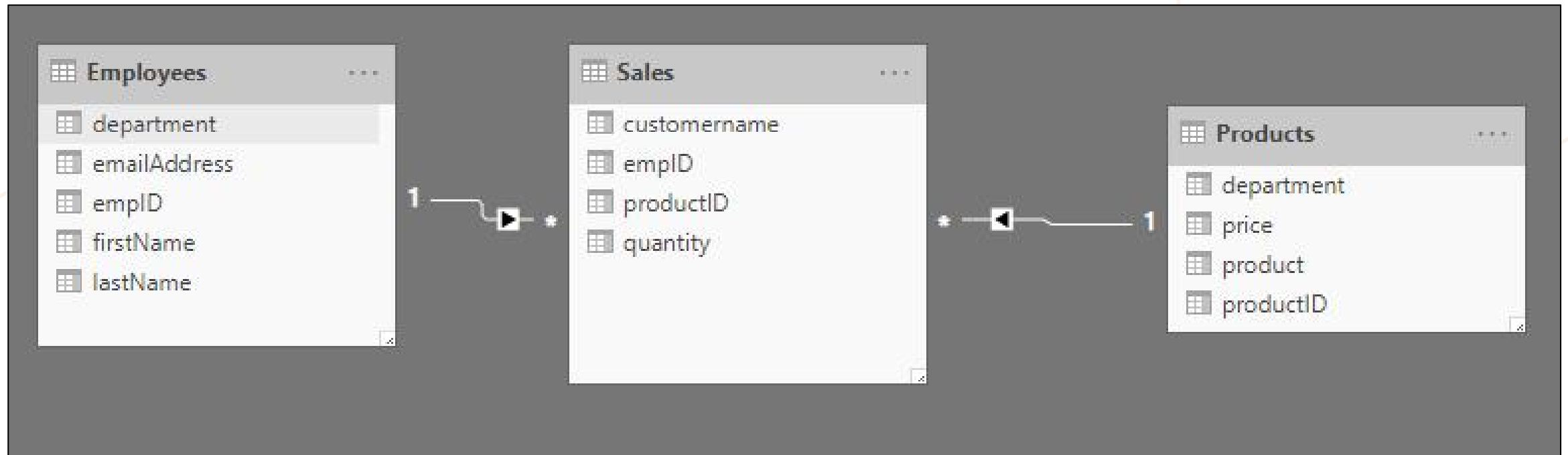
'Components' and 'Bikes' have noticeably more Product.

Implement row-level security





Security overview in Power BI





Configure roles with Static method

Manage roles

Roles

- Automotive ...
- Clothing ...
- Game** ...
- Sports ...

CreateDelete

Tables

- Employees ...
- Sales** ...

Table filter DAX expression

✓✕

[department] = "Game"

Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"

SaveCancel



Use a dynamic value with DAX filters

Manage roles

Roles	Tables	Table filter DAX expression
EmployeeEmailAddress ... CreateDelete	Employees ... Products ... Sales ...	[emailAddress] = userprincipalname()

Explore Copilot for Power BI





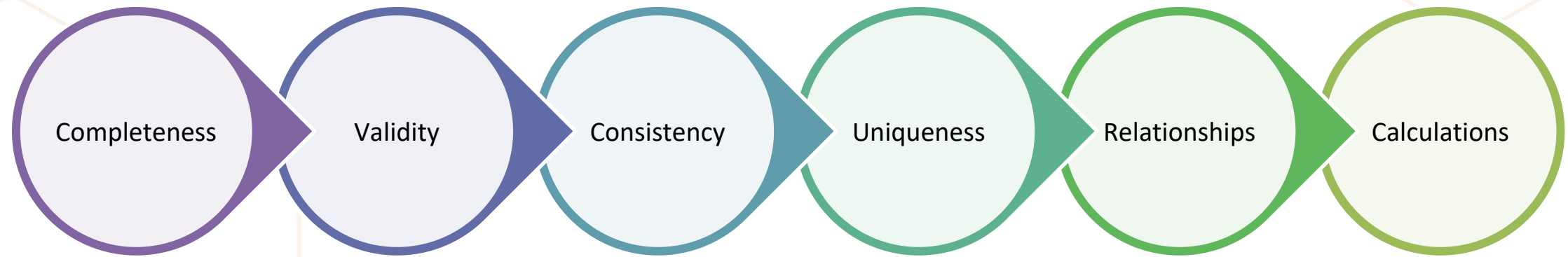
How to use Copilot for Power BI

- Create measures based on natural language.
- Update the semantic model with synonyms for improved user Q&A experience.
- Generate report content, summary visuals, and pages from prepopulated prompts.
- Analyze a summary of underlying semantic model.



Prepare your data and ensure quality

Copilot isn't effective without good data modeling.





Query with DAX

```
// DAX query generated by Fabric Copilot with "total sales for all salespeople individually for all items in the accessories category"
// Filter for Accessories category
EVALUATE
    CALCULATETABLE(
        SUMMARIZECOLUMNS(
            'Salesperson'[Salesperson],
            "Total Sales", SUM('Sales'[Sales])
        ),
        'Product'[Category] = "Accessories"
    )
```

 total sales for all salespeople individually for all items in the accessories category

Explain this query

Explain a DAX topic

Retry

Keep query

Discard query

Content created by AI may contain mistakes, so review it carefully. [Read terms](#)

Salesperson	Total Sales
Stephen Jiang	8374.76
Michael Blythe	38682.84
Linda Mitchell	66916.05

Create measures

- Suggest new measures
- Evaluate returns results
- Update model: Add new measure

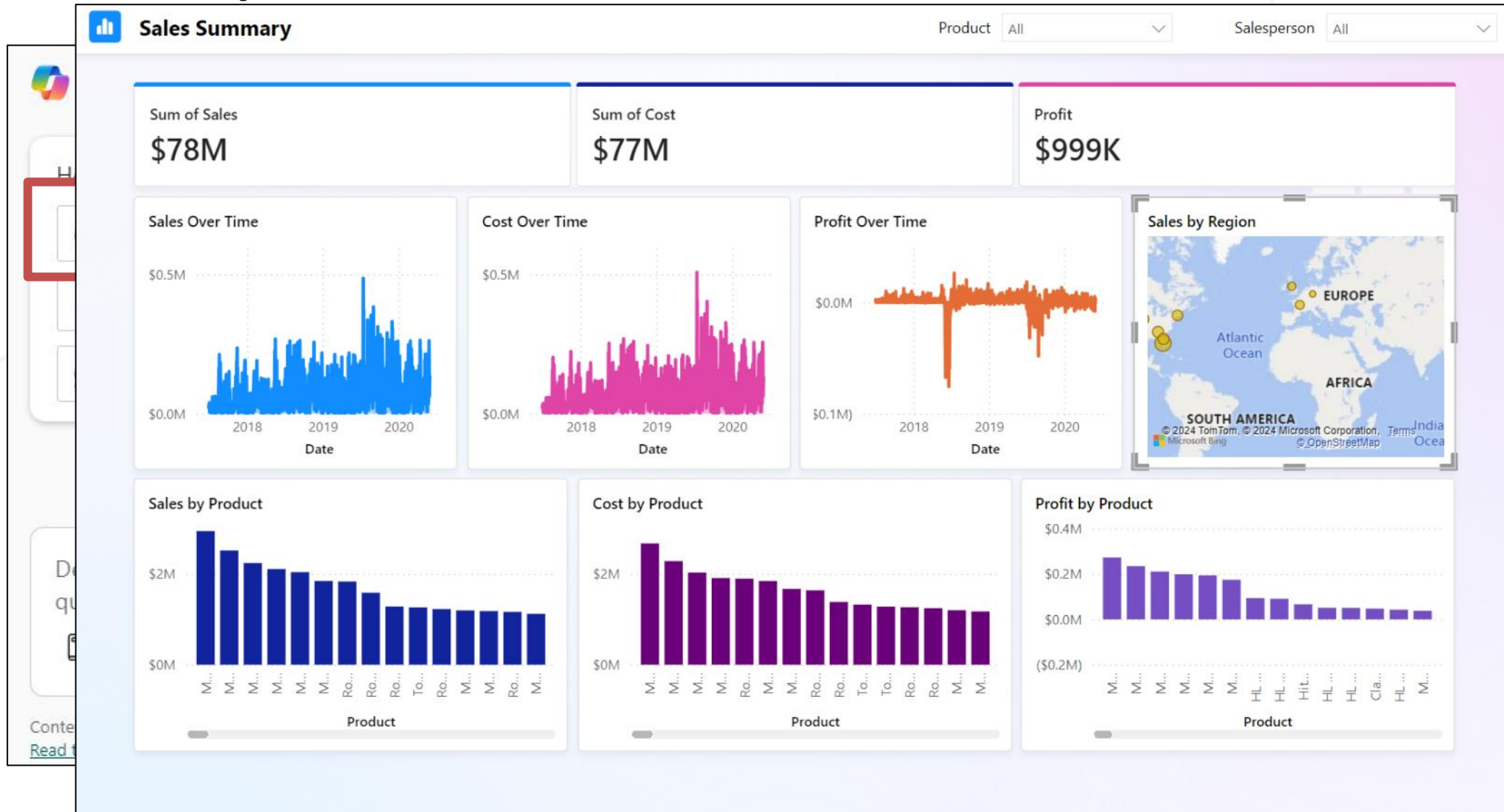
```
1 // DAX query generated by Fabric Copilot with "Suggest new measures in a DAX query for further
2 analysis and try them out with one or more suitable columns"
3 DEFINE
4 // New measure to calculate the average profit per product sold
5 Update model: Add new measure
6 MEASURE 'Sales'[Avg Profit per Product] = DIVIDE([Profit], [Unique Products Sold])
7 // New measure to calculate the average sales per reseller
8 Update model: Add new measure
9 MEASURE 'Sales'[Avg Sales per Reseller] = DIVIDE([Total Sales], [Unique Resellers])
10 // New measure to calculate the average quantity per order
11 Update model: Add new measure
12 MEASURE 'Sales'[Avg Quantity per Order] = DIVIDE([Total Quantity], [Orders])
13 // New measure to calculate the average sales per order
14 Update model: Add new measure
15 MEASURE 'Sales'[Avg Sales per Order] = DIVIDE([Total Sales], [Orders])
16
17 // Evaluate the new measures
18 EVALUATE
19 ROW(
20     "Avg Profit per Product", [Avg Profit per Product],
21     "Avg Sales per Reseller", [Avg Sales per Reseller],
22     "Avg Quantity per Order", [Avg Quantity per Order],
23     "Avg Sales per Order", [Avg Sales per Order]
24 )
```

Results | Result 1 of 1 ▾ | Copy ▾

	[Avg Profit per Product]	[Avg Sales per Reseller]	[Avg Quantity per Order]	[Avg Sales per Order]
1	2992.5	122703.43	56.45	21445.95



Create reports





Suggested and custom prompts

Product Cost and Profitability Analysis

Subcategory: All Category: All

Profit

\$999K

Profit Margin

1.29%

Profit by Product Category

Category	Profit
Components	\$1M
Clothing	\$0.5M
Accessories	\$0.5M
Bikes	\$0.5M

Profit Margin by Product Subcategory

Subcategory	Profit Margin
Tires and Tubes	37.43%
Socks	37.12%
Hydration Packs	36.79%
Cleaners	36.53%
Bottles and Cages	36.36%
Bike Racks	36.26%
Vests	35.61%

Profit Trend Over Time

Accessories Bikes Clothing Components

Profit Margin Trend Over Time

Bib-Shorts Bike Racks Bottles an... Bottom Br... Brakes Caps Chains Cleaners

OK, here's a suggested outline for your report. Select any page topic to view details and start creating pages.

Sales Performance by Salesperson

Regional Sales Analysis

Product Cost and Profitability

Create a page to analyze the cost and profitability of different products, including standard cost, profit, and profit margin.

+ Create

Reseller Performance

Is this response helpful?

Create a page to analyze the cost and profitability of different products, including standard cost, profit, and profit margin.

Created a Product Cost and Profitability Analysis page.

Generate summaries

Create a narrative with Copilot



Describe the summary you want and the visuals it should reference. AI-generated content can

Sales Summary:

Sales by Region:

- The **Southwest region** leads in sales with **\$18M**.
- **Canada** has the second-highest sales at **\$13.88M**.
- **Germany** and **Australia** registered the lowest sales figures at **\$1.88M** and **\$1.39M** respectively ¹.

Sales by Product:

- The **"Mountain-200 Black, 38"** model records the highest sales, totalling **\$2.94M**.
- Significant product sales also include the **"Mountain-200 Black, 42"** and **"Mountain-200 Silver, 38"** models, at **\$2.52M** and **\$2.24M** respectively.
- Lower sales products include the **"LL Road Frame - Red, 52"** and **"Short-Sleeve Classic Jersey, S,"** with sales of **\$20.1K** and **\$45.66K** respectively ².

Sales Over Time:

- A significant spike in sales occurred on **2019-07-15**, totalling **\$486.2K**.
- Another notable sales peak was on **2018-09-13**, with sales reaching **\$262.54K**.
- Sales on **2018-11-27** were lower at **\$136.84K** ³.

Profit:

- The overall **profit** was recorded at **\$999.49K**, with total sales amounting to **\$77.55M** and costs at **\$76.55M** ⁴.

Profit by Product:

- The **"Mountain-200 Black, 38"** generated the highest profit at **\$272.77K**.
- Products such as **"HL Mountain Frame - Silver, 38"** and **"Classic Vest, S"** also recorded notable profits, at **\$93.74K** and **\$47.31K** respectively.
- Products like the **"HL Touring Frame - Blue, 50"** and **"Touring-3000 Blue, 62"** registered losses at **(\$929.15K)** and **(\$9.03K)** ⁵.

Profit Over Time:

- The highest profit on a single day was recorded on **2017-08-01**, at **\$3.65K**.

Created with AI. Inaccuracies are possible. [Read terms](#)

ership

Update



Thank you
Let's discuss our collaborations

PT Nawa Darsana Teknologi
Gedung Office 8, 18th Floor, Jl. Jend Sudirman Kav 52-53
SCBD Lot 8, Senopati Jakarta Selatan 12190 DKI Jakarta,
Indonesia

Website : www.nawatech.co

Phone : +62 21 29552754

Contact

-

-

Email :

Phone :